

LxI 9.1

Answer1:

I chose the **Calc Stats problem**, where the task is to take a sequence of integers and compute the following statistics:

- Minimum value
- Maximum value
- Number of elements
- Average value

I implemented a Calc Stats class with methods such as `minimum()`, `maximum()`, `count()`, and `average()`. The goal was to build these functions step-by-step using Test Driven Development (TDD) and verify each part using `ApprovalTests`.

Answer2:

I solved the problem using **Python 3.14**, following the instructions to use:

- **calstat.py** for the implementation
- **test_calstat.py** for the test cases

All functions were implemented in a beginner-friendly way using loops (without Python shortcuts), matching the TDD workflow.

Answer3:

Working on the Calc Stats problem gave me practical experience with Test Driven Development (TDD). I first wrote tests for each function (`minimum`, `maximum`, `count`, `average`) before implementing the code. This helped me understand exactly what each function should do.

I developed the code incrementally: write a test, see it fail (red phase), write just enough code to pass (green phase), and then refactor if needed. Using `ApprovalTests` taught me that test outputs must match exactly, which made me careful with expected results.

Running `pytest` gave immediate feedback, so I quickly knew which parts needed fixing. By the time all tests passed, I was confident that the logic was correct, there were no hidden bugs, and

every function worked as expected. Overall, TDD made the development structured, controlled, and reliable.